Implicit regularization

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Definition

Ω: set of parameters; $L(\theta)$: population loss of a neural network $f(\theta, x)$ with input data x.

Gradient descent:

$$\Delta\theta_k = -\eta \nabla L(\theta_k)$$

Convergence to $\theta^\star \in \Omega^\star = \mathrm{argmin}_\theta L(\theta)$

Implicit regularization: there exists a complexity function $R[\theta]$:

$$\theta^\star = \min_{\theta \in \Omega^\star} R[\theta]$$

GD prefers solutions within certain complexity classes

Explicit regularization

Define regularized loss and optimize it:

$$\mathcal{L}(\theta, x) = L(\theta, x) + \lambda R(\theta)$$

Example weight decay:

$$R = ||\theta||_2^2$$

Why do we care?

- Contrary to naive statistics, deep neural networks (DNNs) generalize surprisingly well
- Implicit regularization must occur
- SLT tells us a story for Bayesian learning
- We want a theory of implicit regularization for SGD
- Implicit regularization from scale (depth), loss, optimizer, initialization,...

Implications for alignment

- Alignment is also a question about implicit regularization
- Observe policy $\pi(\theta, x)$: the policy does well on the reward and looks aligned in behavioural tests
- But perhaps $\pi(\theta, x)$ is scheming (misaligned) because it is favoured by training
- If there is no strong regularization for honest solutions, then schemers are more likely
- Hope 1: Understanding implicit bias helps us detect scheming
- Hope 2: Design a training process that selects against schemers

Loss landscape geometry

Critical points

- They give the menu of possible solutions
- They shape the training dynamics
- $\nabla L(\theta) = 0$

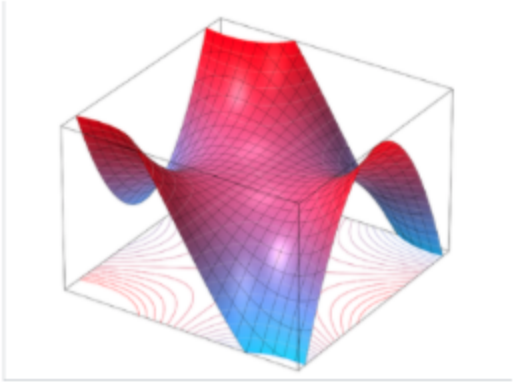


Figure 1: Example of a landscape with a plateau (non-strict saddle point).

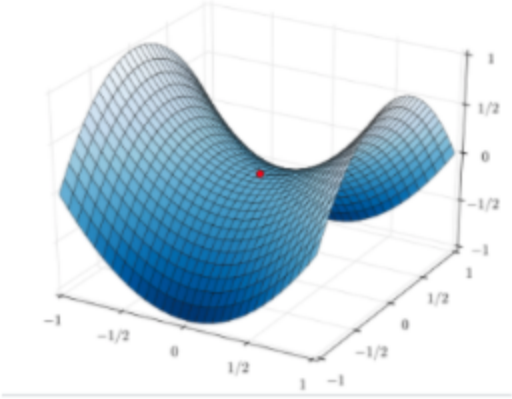


Figure 2: Example of a landscape with a strict saddle point at (0,0).

Deep Linear Networks

Function (multilinear in parameters):

$$f: \mathbb{R}^{d_0} \times \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_L}$$

$$(x, W_1, \ldots, W_L) \mapsto W_L \ldots W_1 x = Wx$$

Function W (student) learns M (teacher)

We train on data generated by $y = Mx$ with quadratic loss and input covariance matrix $\Sigma_X = I$:

$$\mathcal{L}_N(\theta; D_N) = \frac{1}{2N} \sum_{i=1}^{N} ||y_i - W_L \ldots W_1 x_i||_2^2 = \frac{1}{2N} \sum_{i=1}^{N} ||Mx_i - Wx_i||_2^2 \to_N ||(M - W)||_F^2$$

Non-convex, high-dimensional loss landscape

Symmetries

Let $GL_{\mathbf{d}} := \Pi_{l=L}^{0} GL_{d_l} = GL_{d_L} \times GL_h \times GL_{d_0}$ act on Ω via:

$$\theta = (W_1, \ldots, W_L) \mapsto (g_1 W_1 g_0^{-1},\ g_2 W_2 g_1^{-1},\ \ldots,\ g_L W_L g_{L-1}^{-1})$$

Define $\mu(\theta) = W$

$$\mu(g \cdot \theta) = g_L W g_0^{-1}; \quad \mu(g_h \cdot \theta) = \mu(\theta)$$

Let $\theta \in \pi^{-1}(W)$, the orbit $\mathcal{O}_\theta := G_{\mathrm{h}} \cdot \theta$ is a subspace of the fiber $\mu^{-1}(W)$

Global minima

- If $d_l \geq d_L$, there is a unique solution in function space for $W^\star = \Sigma_{YX} = M$ (OLS), with Σ_{YX} the input-output covariance
- A whole space of parameters maps to the solution (degenerate)
- Key geometric invariants under symmetries: rank patterns $r_{ij} = \mathrm{rank}(W_i \dots W_j)$
- Can stratify the degenerate parameters mapping to global minima into $O_{r_{ij}}$
- Gives a combinatorial description of global minima
- Can be used to compute the dimension of global minima (link with RLCT)
- See more at this paper from Simon Pepin Lehalleur and Richárd Rimányi
- Informally: the rank patterns quantify simplicity among degenerate solutions

Saddle points

In DLNs, if $d_l \geq d_L$, with full-rank and non-degenerate data:

- **All critical points are either saddles or global minima**
- Saddles can be strict or non-strict
- They are also often highly degenerate (many flat directions)
- This is generic in DNNs

Work is in progress to get the combinatorics of saddles with quiver representation theory

First order critical points

Assume $d_l \geq d_L$ and well-behaved data. Let θ be a critical point of the population loss $\mathcal{L}$.

From Achour et al., there exists a subset S indexing the left singular vectors of:

$$W^\star = M = U\mathrm{diag}(s_1, \dots s_{d_L}, 0, \dots, 0)V^\top$$

such that for $P_S := U_S U_S^\top$ a linear projector onto the space of vectors indexed by the set S:

$$\mu(\theta) := W = W_L \dots W_1 = P_S W^\star = U_S U_S^\top \Sigma_{YX} \Sigma_X^{-1}$$

Takeaways

- Loss landscapes are highly degenerate
- Under some conditions: local minima are global
- Discover geometric invariants by studying their symmetries
- A world of topology, geometry, group representation theory

Learning Rate

Descent lemma and stability

If $\mathcal{L}$ is convex and σ-smooth, then for gradient descent with any $\eta < \frac{2}{L}$,

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta \left(1 - \frac{\eta\sigma}{2}\right) \|\nabla \mathcal{L}(\theta_t)\|_2^2$$

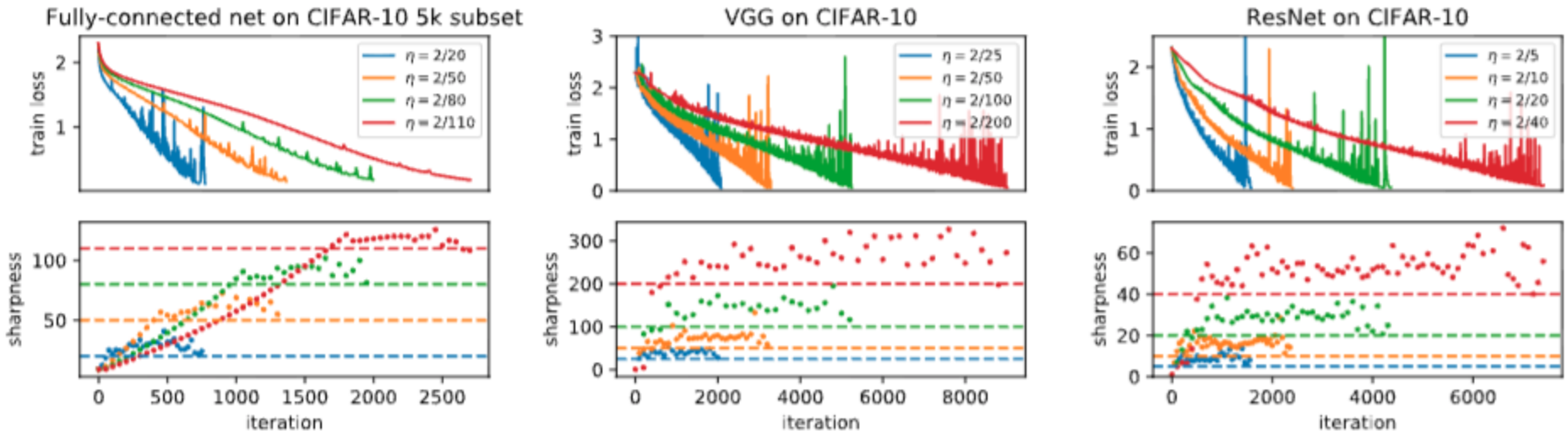
Local stability near a minimum is controlled by the largest Hessian eigenvalue:

$$\sigma \approx \lambda_{\max}(H) \qquad \Rightarrow \qquad \eta < \frac{2}{\lambda_{\max}(H)}$$

Stability threshold ensuring monotonic decrease of the loss in convex optimization

Edge of stability

The loss is non-convex, but the Hessian oscillates slightly above $\frac{2}{\eta}$: it regularizes away sharp minima



Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, Cohen et al.

Gradient flow

Gradient flow and NTK dynamics

Small-learning-rate limit of gradient descent:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \qquad \rightsquigarrow \qquad \dot{\theta}(t) = -\nabla_\theta \mathcal{L}(\theta(t))$$

Neural Tangent Kernel (NTK):

$$K(x, x', \theta) = \nabla_\theta f(\theta, x) \nabla_\theta f^\top(\theta, x) = \sum_{i=1}^{p} \partial_{\theta_i} f(\theta, x) \partial_{\theta_i} f^\top(\theta, x)$$

For a quadratic loss on a dataset (x_i, y_i), the dynamics in function space are

$$\frac{d}{dt} f(\theta, x) = -\sum_{i=1}^{N} K_\theta(x, x_i) \big(f_\theta(x_i, t) - y_i \big)$$

Conserved quantities

Conserved quantities under gradient flow:

$$G_l = W_{l+1}^\top W_{l+1} - W_l W_l^\top$$

When $G_l = 0$ we have **balanced weights**

For a 1D, 2-layer DLN, we get a hyperbolic equation: $w_2^2 - w_1^2 = G$

See the tutorial for more

Self-consistent equation in DLNs

Balanced weights $\implies$ self-consistent gradient-flow equation in function space:

$$\dot{W} = \sum_{k=1}^{L} (WW^\top)^{\frac{L-k}{L}} \nabla_W \mathcal{L}(W) (W^\top W)^{\frac{k-1}{L}}$$

With operator $K : Q \mapsto \sum_{k=1}^{L} (WW^\top)^{\frac{L-k}{L}} Q (W^\top W)^{\frac{k-1}{L}}$

$$\dot{W} = K[\nabla \mathcal{L}(W)]$$

Heuristic implicit regularization

- Consider balanced weights of rank-r students.
- Recall $\mu^{-1}(W)$ has many symmetries: rank patterns
- The singular values of weight W^l are σ_i^l, and $p_k = \frac{\sigma_k^l}{\sum_k \sigma_k^l}$
- Spectral Entropy: $H^l(p) = -\sum_k p_k \log(p_k)$
- Entropy $S(W) = \sum_{l=1}^{L} H^l(p)$
- Free energy: $F(W) := L(W) - \frac{1}{\beta} S(W)$ with inverse temperature β
- Recall GF is $\dot{W} = -K[\nabla L(W)]$
- GF effectively minimizes a free energy functional, and entropy makes the regularization explicit
- For more, see The geometry of the deep linear network, Govind Menon

Initialization

Lazy Regime (NTK regime)

For a DLN of depth L with initialization $\theta(0) \sim N(0, \sigma)$ and $\sigma^2 = d_L^{-\gamma},\ \gamma < 1$:

$$\dot{W} \approx K_0[M - W]$$

Leading to the following solution:

$$W(t) = M - \exp(-K_0 t)[M - W(0)]$$

The time scale of learning is fixed by the NTK at initialization:

$$t \approx -k_0^{-1} \log\left(\frac{s - w_f}{s - w_0}\right)$$

Behaves like a linear network with no simplicity bias

Rich regime

For balanced and aligned weights (or $\gamma > 1$), GF reduces to a system of 1D logistic ODEs:

$$\dot{w}_\alpha = 2\left(s_\alpha - w_\alpha\right) w_\alpha.$$

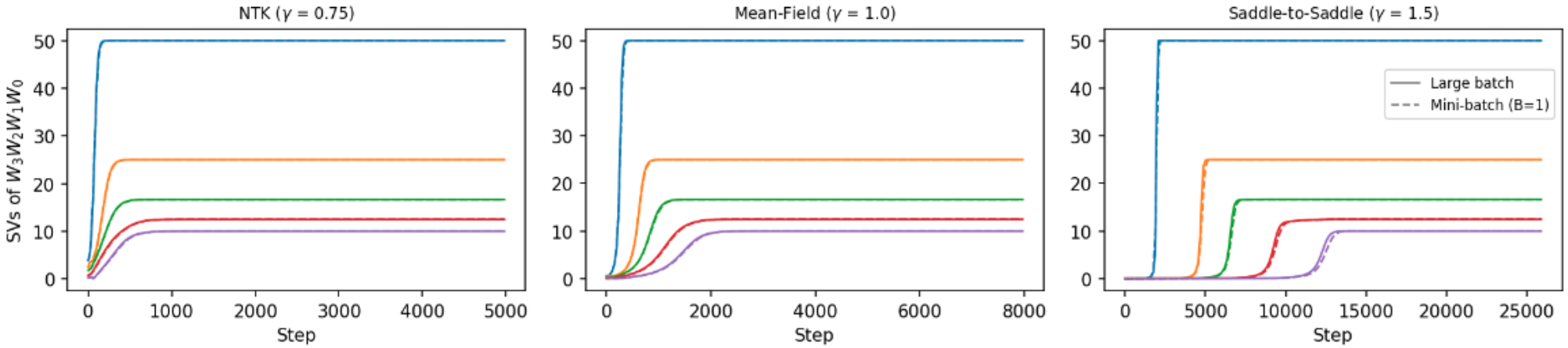
Solving this ODE for $w(t) := w_\alpha$ gives:

$$w(t) = \frac{s}{1 + \left(\frac{s}{w_0} - 1\right) e^{-2st}}.$$

Timescale of learning

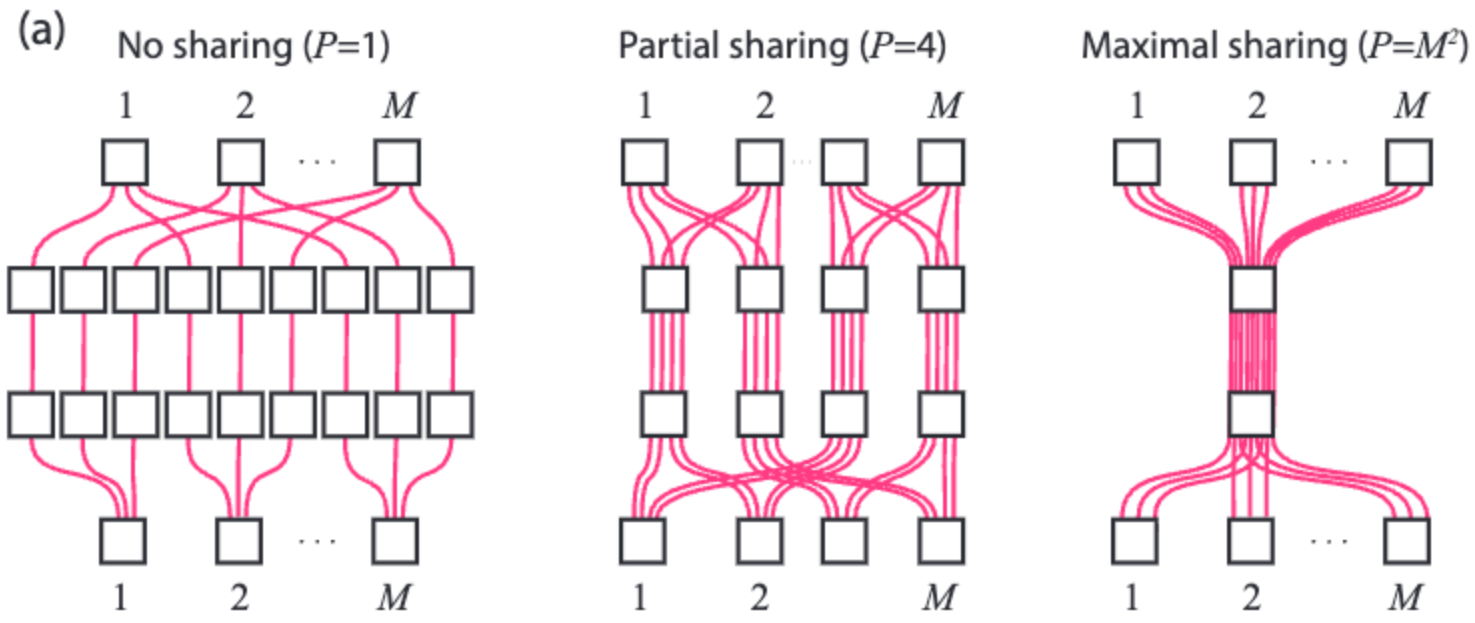
$$t = \frac{1}{2s} \ln\left(\frac{w_f\left(s - w_0\right)}{w_0\left(s - w_f\right)}\right), \qquad s \neq 0.$$

Regimes


NTK (γ = 0.75)
Mean-Field (γ = 1.0)
Saddle-to-Saddle (γ = 1.5)
SVs of $W_3W_2W_1W_0$
Step
Large batch
Mini-batch (B=1)

Other architectures

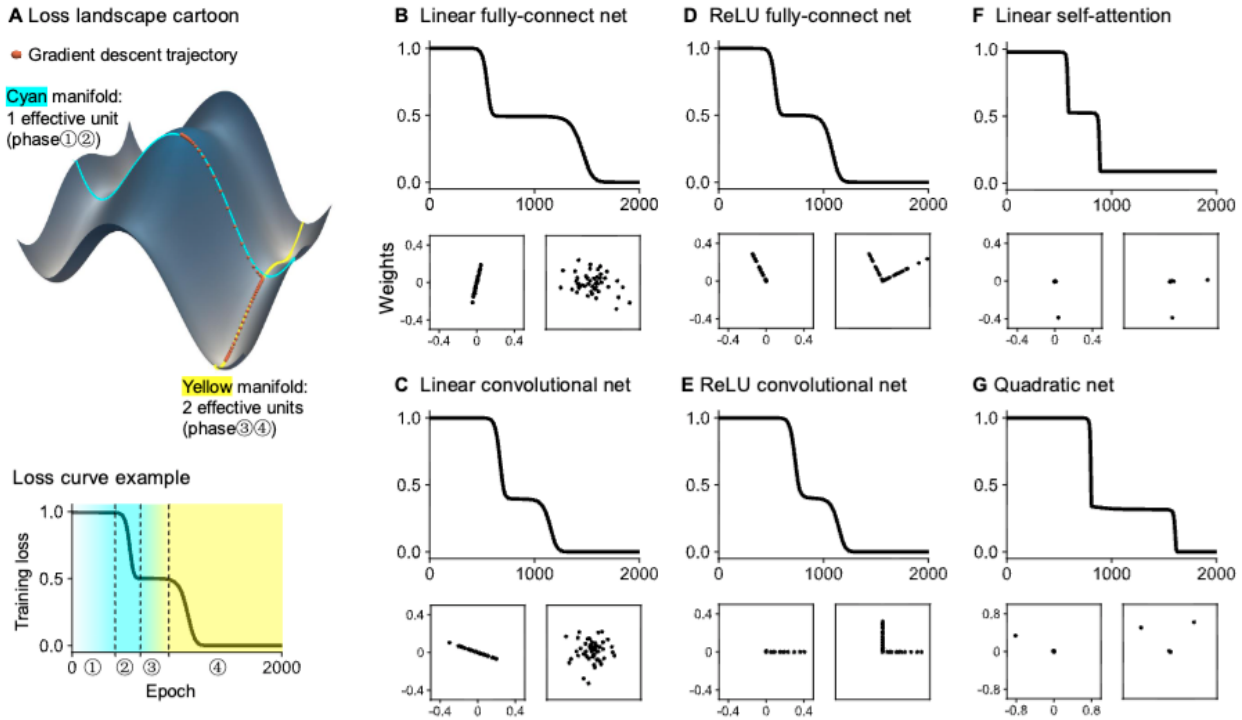
Nonlinearity: neural race reduction



The Neural Race Reduction: Dynamics of Abstraction in Gated Networks, Andrew M. Saxe et al.

MLP and linear transformers

Saddle-to-Saddle Dynamics Explains A Simplicity Bias Across Neural Network Architectures, Saxe et al.



Stochasticity

Langevin dynamics

- SGD: Sample a batch from a dataset and compute the batch gradient
- Continuous model with B_t Brownian

$$d\theta_t = -\nabla L(\theta_t)\, dt + \sqrt{\eta\, \Sigma(\theta_t)}\, dB_t,$$

- Gradient noise covariance $\Sigma(\theta_t)$ depends on geometry: it is anisotropic and state-dependent

Fokker-Planck: Boltzmann equilibria

The Fokker-Planck equation (for density $p(\theta, t)$) is a local conservation law:

$$\begin{aligned} \partial_t p &:= -\nabla \cdot j \\ j &:= -\nabla L(\theta) p(\theta) - \frac{\eta}{2} \nabla^\top \left(\Sigma(\theta) p(\theta) \right) \end{aligned}$$

Important special case:

- Stationary distribution: $\partial_t p^*(x, t) = 0$
- Thermal equilibrium: $j = 0$
- White noise: $\Sigma = \sigma^2 I$

$$p^*(\theta) \propto \exp\left(-\frac{2}{\eta \sigma^2} L_N(\theta) \right) \text{ Boltzmann}$$

Noise anisotropy

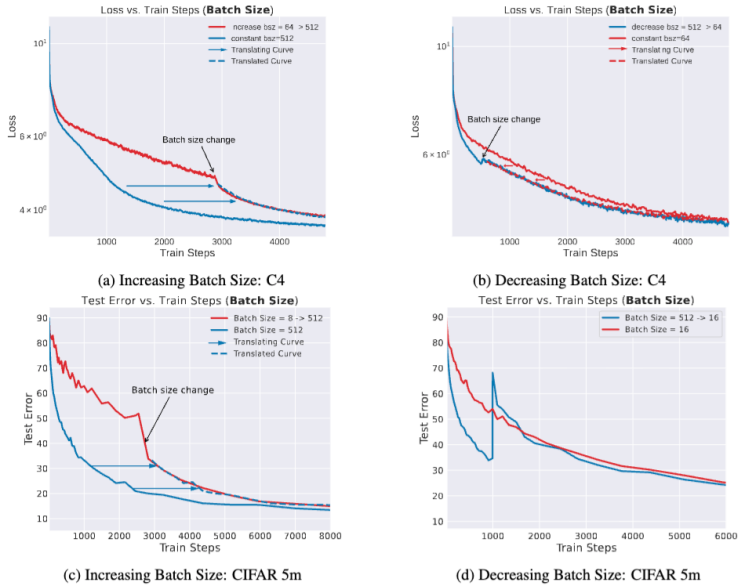
- In general, noise is anisotropic: it introduces a different implicit bias
- In function space, Langevin introduces a drift

$$\frac{\eta}{2}\mathrm{Tr}(\Sigma(\theta)H(\theta))$$

- Heuristic: bias toward flatter minima
- More:
  - see Govind Menon On the implicit regularization of Langevin dynamics with projected noise
  - My paper with collaborators deriving Langevin on DLNs during the saddle-to-saddle regime

Golden Path hypothesis

- Lots of debate about the implicit bias of SGD noise
- Transient (drift dominated) vs low-loss regime (diffusion dominated):



(a) Increasing Batch Size: C4 (b) Decreasing Batch Size: C4

(c) Increasing Batch Size: CIFAR 5m (d) Decreasing Batch Size: CIFAR 5m

Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

Discussion

- Implicit bias: making explicit the class of solutions that the training process selects
- Different causes of implicit regularization
- Often there is a simplicity bias
- How strong it is matters for alignment
- Many open questions
    - Mostly, we understand DLNs, but not fully
    - Beyond DLNs (gated DLNs, ReLU)
    - Structure of the data (important)